N8n Workflow Automation

Introduction

** **n8n is a workflow automation tool** that helps us automate repetitive tasks by connecting different applications and services

** **Low-code platform**: We can build workflows visually by connecting nodes, with minimal coding.

** **Workflow automation:** We can automate tasks such as sending emails, saving data, processing files, and updating databases.

** **Node-based architecture:** Each node performs a specific task, such as receiving input, calling an API, or processing data.

** **Trigger-based execution:** A workflow can start when a user sends a message, a webhook receives a request, or a scheduled time arrives.

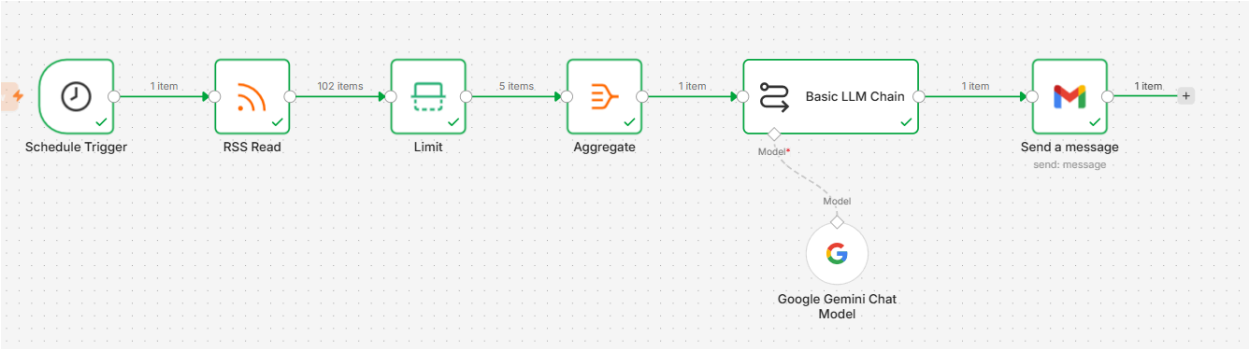
** **AI integration:** We can connect n8n with LLMs such as OpenAI models to build AI-powered applications and chatbots.

** **Application integration:** It connects services such as Google Sheets, Gmail, databases, APIs, and messaging platforms.

In one sentence: n8n helps us connect applications, automate workflows, and build AI agents without writing an entire application from scratch.

Building AI News Agent using n8n

Objective : Build an AI News Agent that automatically collects AI news articles, selects the five most important stories using an LLM, generates an HTML newsletter, and sends it to our Gmail account.



Step-1 : Create New Workflow

- ➔ Open your n8n instance.
- ➔ Click Create Workflow.
- ➔ Set the workflow name to AI News Agent.

Step-2 : Add Schedule Trigger

- ➔ Add a Schedule Trigger node.
- ➔ Configure it to run daily.
- ➔ Select your preferred time.
- ➔ For example, run it every day at 8:00 AM.

Step-3 : Add RSS Read

- ➔ Add an RSS Read node.
- ➔ Connect it to the Schedule Trigger.
- ➔ Configure the RSS Feed URL:

https://news.google.com/rss/search?q=Artificial+Intelligence

Note : This node retrieves recent AI-related news articles from Google News RSS.

Step-4 : Add Limit

- ➔ Add a Limit node.
- ➔ Connect it to RSS Read.
- ➔ Set the maximum number of items to 5.

Note: This limits the input to five articles. If you want the LLM to select the five most important stories from a larger collection, configure the RSS source to retrieve more articles and apply the Limit after the LLM or to the LLM's selected results, as appropriate.

Step-5 : Add Aggregator with All items data into single item

- ➔ Add an Aggregate node.
- ➔ Configure it to collect the article data into a single item.
- ➔ Include useful fields such as:
    - Article title
    - Article link
    - Publication date
    - Article description

The purpose is to send multiple news articles together to the LLM.

Step-6 : Add Basic LLM Chain and configure below prompt with Source for prompt as 'define below' option.

- ➔ Add a Basic LLM Chain node.
- ➔ Connect the Aggregate node to it.
- ➔ Set the prompt source to Define Below.
- ➔ Paste the following prompt.

---------- prompt start ----------

You are an AI News Editor.
Read the following AI news articles.
Select the TOP 5 most important news.

IMPORTANT:
Return ONLY valid HTML.
Do NOT use Markdown.
Do NOT wrap the HTML in ```html or ```.
Use this exact structure:

<h1>📰 Today's Top 5 AI News</h1>
For each news:

<h2>News Title</h2>
Summary:
Two line summary here.
Why it matters:
Explain why this news is important.
Read Full Article →

Finally add:
<hr>
<h2>🔥 Today's Biggest AI Trend</h2>
Trend explanation here.
News:
{{ JSON.stringify($json.data) }}

---------- prompt end ----------

Step 7: Add Google Gemini Chat Model

- ➔ Add a Google Gemini Chat Model node.
- ➔ Connect it to the Basic LLM Chain through its model connection.
- ➔ Configure your Google Gemini credentials or API key.
- ➔ Select an available Gemini model.
- ➔ The LLM reads the collected articles, evaluates their importance, and generates the HTML newsletter.

  GENERATE KEY HERE : https://aistudio.google.com/api-keys

Step-8 : Configure Gmail with Send Message option (Use Google Sign in option)

- ➔ Add a Gmail node.
- ➔ Select the Send a Message operation.
- ➔ Connect it after the Basic LLM Chain.
- ➔ Sign in using Google authentication.
- ➔ Configure the recipient email address.

Gmail Subject :: Today AI News | {{$now.toFormat("dd MMM yyyy")}}

Gmail Body :: {{$json.text}}

Step 9: Test the Workflow

- ➔ Execute the RSS Read node and confirm that articles are retrieved.
- ➔ Verify that the Aggregate node combines the article data correctly.
- ➔ Execute the Basic LLM Chain and inspect the generated HTML.
- ➔ Execute the Gmail node and verify that the email arrives with properly formatted content.
- ➔ Activate the workflow to enable scheduled execution.

n8n Workflow

Schedule Trigger
↓
RSS Read
↓
Limit (5)
↓
Aggregate
↓
Basic LLM Chain
↓
Google Gemini Chat Model
↓
Send Email using Gmail

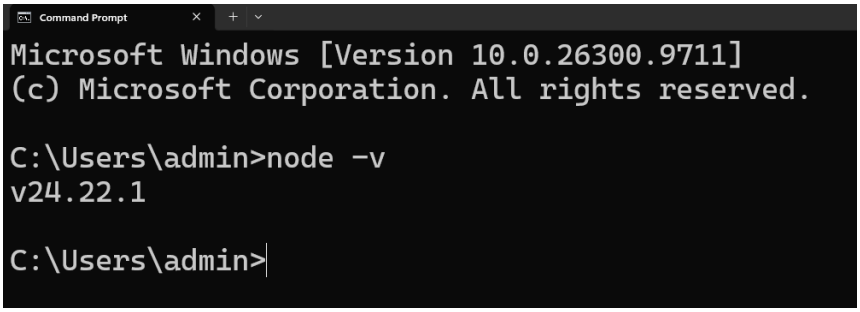
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n8n Setup in local

Step 1: Download and Install Node.js

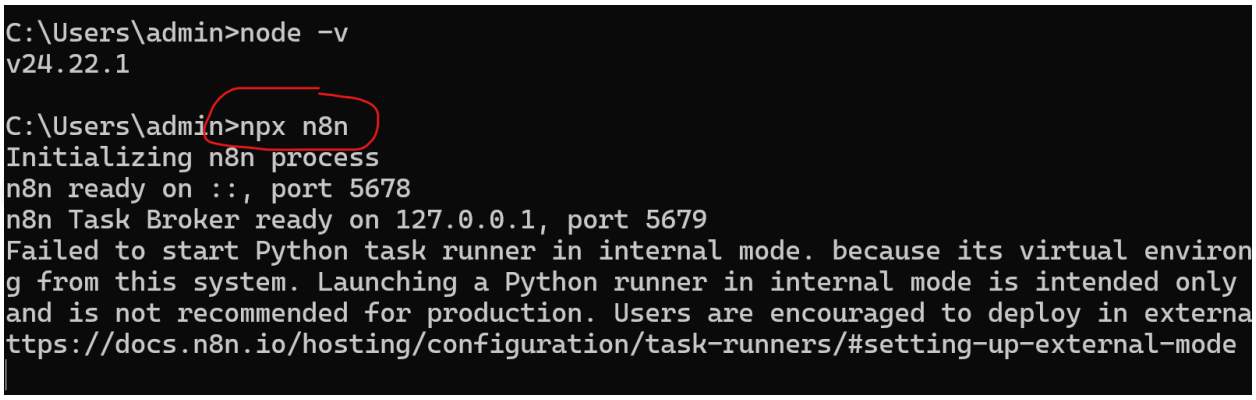
- Open the official website: https://nodejs.org/
- Download the **LTS (Long-Term Support)** version.
- Run the installer and complete the installation.
- Node.js includes npm (Node Package Manager).

Step 2 : Verify Node version



Step-3 : Install n8n using below command

- npx runs a Node.js package without requiring a global installation.
- n8n starts the n8n workflow automation platform.
- If prompted to download n8n, confirm the installation.

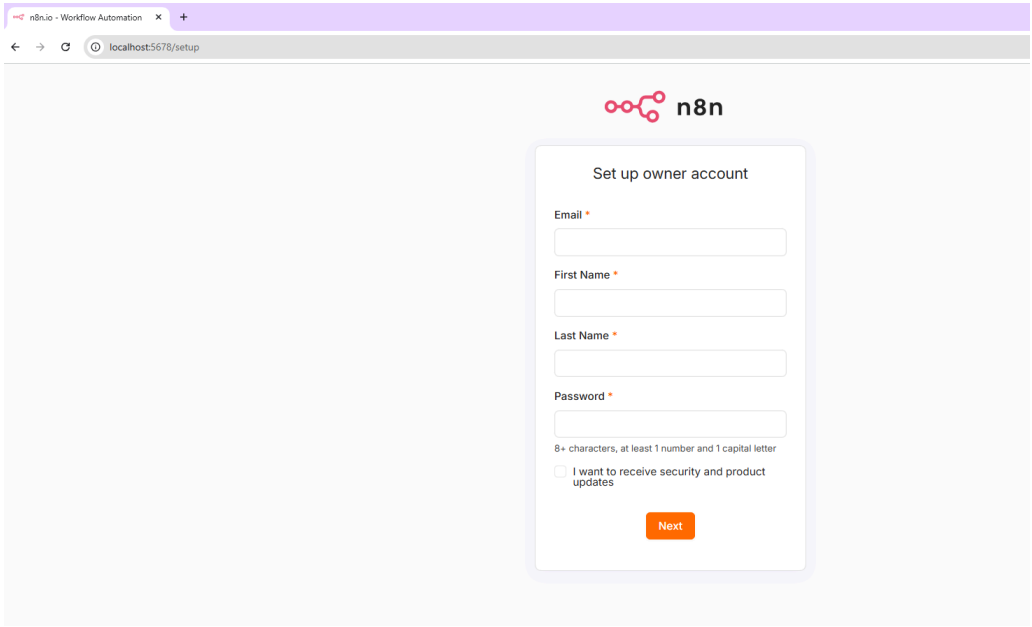


Step 4: Open n8n in the Browser

Once n8n starts successfully, open:

http://localhost:5678

Create your local account if prompted and access the n8n workflow editor.



Assignment

Using n8n workflow create an AI Agent to find the latest jobs from linkedin for GEN AI Skills and send to gmail everyday at 9:00 PM IST.

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